

1.6.2 Proof of the valence formula

The proof presented here avoids algebraic geometry, at the cost of some complex analysis. Let f be a modular function, and consider the following contour:

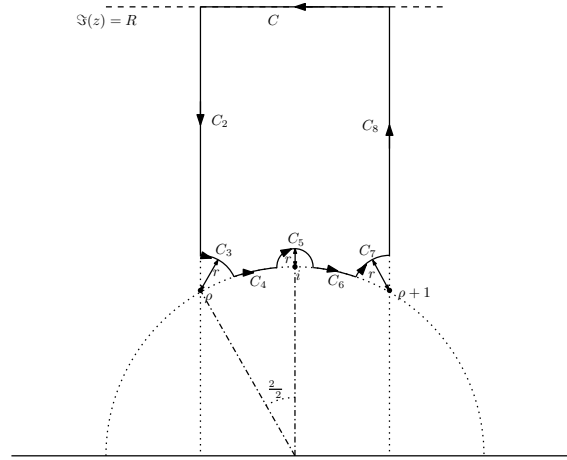


Figure 1.3: Contour to integrate

The proof consists in integrating $f'(z)/f(z)$ around the indicated contour C in two different ways.

Integral residue formula:

This is the first way in which we calculate the contour integral. For R large enough so that the contour contains all zeros and poles of $\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}'$ with large imaginary part, we have:

$$\int_C \frac{f'(z)}{f(z)} dz = 2\pi i \sum_{p \in \text{int}(C)} \text{Res}_p \left(\frac{f'}{f} \right) = 2\pi i \sum_{p \in \mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}'} v_p(f).$$

The rest of the proof consists in computing the integral as the sum of the path integrals along $C_1, \dots, C_8$.

Integral along C_1 :

Perform the change of variables $q(z) = e^{2\pi iz}$. Note that:

$$dq = 2\pi i q dz, \quad dz = \frac{1}{2\pi i} \frac{dq}{q}.$$

Moreover, the path $q(C_1)$ is a clockwise circle of radius $e^{-2\pi R}$ centered around 0. Finally,

$$\frac{d}{dz} f(q) = \frac{d}{dz} f(e^{2\pi iz}) = 2\pi i q f'(q).$$

Putting these together, we obtain:

$$\int_{C_1} \frac{f'(z)}{f(z)} dz = \int_{q(C_1)} \frac{f'(q)}{f(q)} dq.$$

Since f is meromorphic around infinity, this last quantity equals $v_\infty(f)$.

Integral along C_2 and C_8 :

Note that since f is modular it satisfies $f(z+1) = f(z)$. Therefore also $f'(z+1) = f'(z)$, and we get, by changing $w = z+1$:

$$\int_{C_8} \frac{f'(w)}{f(w)} dw = \int_{C_2} \frac{f'(z+1)}{f(z+1)} dz = \int_{C_2} \frac{f'(z)}{f(z)} dz.$$

Therefore the contributions of C_2 and C_8 cancel out.

Integral along C_4 and C_6 :

In the same way that for C_2 and C_8 we considered the change $z \mapsto z + 1$, here we consider the change $z \mapsto s(z) = z^{-1}$. This change of variables transforms C_4 to C_6 . Weak-modularity of f implies also that $f(z) = z^{-k} f(s(z))$. Differentiating both sides yields:

$$f'(z) = -kz^{-k-1}f(s(z)) + z^{-k}f'(s(z))s'(z).$$

This gives

$$\frac{f'(z)}{f(z)} = -\frac{k}{z} + \frac{f'(s(z))s'(z)}{f(s(z))}.$$

Therefore we may compute

$$\int_{C_4} \frac{f'(z)}{f(z)} dz = \int_{C_4} -\frac{k}{z} dz + \int_{C_4} \frac{f'(s(z))s'(z)}{f(s(z))} dz = 2\pi i \frac{k}{12} + \int_{C_6} \frac{f'(s)}{f(s)} ds,$$

and hence

$$\int_{C_4+C_6} \frac{f'(z)}{f(z)} dz \xrightarrow{r \rightarrow 0} 2\pi i \frac{k}{12}.$$

Integral along C_5 :

Using the argument principle we obtain

$$\int_{C_5} \frac{f'(z)}{f(z)} dz \xrightarrow{r \rightarrow 0} \frac{1}{2} 2\pi i v_i(f).$$

Integrals along C_3 and C_7 :

Applying again the argument principle we obtain

$$\int_{C_3} \frac{f'(z)}{f(z)} dz \xrightarrow{r \rightarrow 0} \frac{1}{6} 2\pi i v_\rho(f), \text{ and } \int_{C_5} \frac{f'(z)}{f(z)} dz \xrightarrow{r \rightarrow 0} \frac{1}{6} 2\pi i v_{\rho+1}(f) = \frac{1}{6} 2\pi i v_\rho(f).$$

Therefore

$$\int_{C_3+C_5} \frac{f'(z)}{f(z)} dz \xrightarrow{r \rightarrow 0} \frac{1}{3} 2\pi i v_\rho(f).$$

Conclusion:

Putting together all the above calculations yields the sought formula.

1.7 product formula for $\zeta(z)$

Consider the weight-2 Eisenstein series

$$G_2(z) = \sum_c \sum'_d \frac{1}{(cz+d)^2},$$

which does not converge absolutely. It is *not* a weight-2 modular form (there are no such other than 0), but we still have the formula:

$$G_2(z) = 2\zeta(2)E_2(z), \quad E_2(z) = 1 - 24 \sum_{n=1}^{\infty} \sigma_1(n)q^n.$$

The function $E_2(z)$ is holomorphic on $\mathbb{H}$, and $E_2(z+1) = E_2(z)$. However, we are about to see that

$$z^{-2} E_2(-1/z) = E_2(z) + \frac{12}{2\pi i z}.$$

Sometimes these functions are called *quasi-modular*.

Theorem 1.7.1 The function G_2 satisfies $G_2(z+1) = G_2(z)$ and it has the q -expansion

$$G_2(z) = 2\zeta(2) - 8\pi^2 \sum_{n=1}^{\infty} \sigma_1(n)q^n, \quad q = e^{2\pi iz}.$$

Moreover, G_2 satisfies the transformation property:

$$G_2(\gamma z) = (cz+d)^2 G_2(z) - 2\pi ic(cz+d), \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \quad (1.7)$$

In particular, the non-holomorphic function $G_2^*(z) = G_2(z) - \pi/\Im(z)$ is weight-2-invariant for $\mathrm{SL}_2(\mathbb{Z})$.

Proof. To show that $G_2(z+1) = G_2(z)$, we must show that

$$\sum_{n \neq 0} \frac{1}{(m(z+1)+n)^2} = \sum_{n \neq 0} \frac{1}{(mz+n)^2}.$$

This follows from the fact that the sum converges absolutely and $n \mapsto n+m$ is a bijection of $\mathbb{Z}$ (for each fixed $m \in \mathbb{Z}$).

Next, we compute the q -expansion of G_2 . First, note that

$$G_2(z) = 2\zeta(2) + 2 \sum_{m=1}^{\infty} \sum_{n \in \mathbb{Z}} \frac{1}{(mz+n)^2}.$$

Using now Lemma 1.4.5 with z substituted by mz , we obtain

$$G_2(z) = 2\zeta(2) - 8\pi^2 \sum_{m=1}^{\infty} \sum_{d=1}^{\infty} dq^{md}.$$

Grouping terms contributing to a given power of q gives the formula.

Next, by expanding $G_2(\gamma_1 \gamma_2 z)$, using the cocycle property of $j(\gamma, z)$ and calculating the lower left entry of the product of two matrices, we see that if Equation 1.7 is satisfied for matrices γ_1 and γ_2 then it is also satisfied for $\gamma_1 \gamma_2$ and γ_1^{-1} . Therefore to prove Equation 1.7 it will be enough to prove it for the matrix $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

We next show that

$$G_2(-1/z) = z^2 \left(2\zeta(2) + \sum_{n \in \mathbb{Z}} \sum_{m \neq 0} \frac{1}{(mz+n)^2} \right).$$

This only differs from $z^2 G_2(z)$ in the order of summation! This is done by substituting $-1/z$ in the definition to get:

$$G_2(-1/z) = \sum_{n \neq 0} \frac{1}{n^2} + \sum_{m \neq 0} \sum_{n \in \mathbb{Z}} \frac{z^2}{(mz - n)^2} = \sum_{n \neq 0} \frac{1}{n^2} + \sum_{m \neq 0} \sum_{n \in \mathbb{Z}} \frac{z^2}{(m - nz)^2}$$

This can be rewritten as

$$2\zeta(2) + z^2 \left(\sum_{n \neq 0} \sum_{m \neq 0} \frac{1}{(mz+n)^2} \right) = 2\zeta(2) + z^2 \left(\sum_{n \neq 0} \sum_{m \neq 0} \frac{1}{(mz+n)^2} + 2\zeta(2) \right).$$

The outer term $2\zeta(2)$ can be put into the sum corresponding to the term $n = 0$, getting the desired formula. The partial fraction decomposition

$$\frac{1}{(mz+n)(mz+n+1)} = \frac{1}{mz+n} - \frac{1}{mz+n+1}$$

allows us to show (via a telescoping sum) that

$$\sum_{m \neq 0} \sum_{n \in \mathbb{Z}} \frac{1}{(mz+n)(mz+n+1)} = 0$$

Subtracting this series from the definition of $G_2(z)$ (which can be done term by term, this only needs conditional convergence) we get the new formula involving an absolutely convergent sum:

$$G_2(z) = 2\zeta(2) + \sum_{\substack{m \neq 0 \\ n \in \mathbb{Z}}} \frac{1}{(mz+n)^2(mz+n+1)}.$$

Then we compute

$$z^{-2}G_2(-1/z) - G_2(z) = \sum_{n \in \mathbb{Z}} \sum_{m \neq 0} \frac{1}{(mz+n)^2} - \sum_{m \neq 0, n \in \mathbb{Z}} \frac{1}{(mz+n)^2(mz+n+1)}.$$

Subtracting term by term and using that

$$\frac{1}{(mz+n)^2} - \frac{1}{(mz+n)^2(mz+n+1)} = \frac{1}{(mz+n)(mz+n+1)}$$

gives an alternative formula for $G_2(z)$:

$$G_2(z) = z^{-2}G_2(-1/z) - \sum_{n \in \mathbb{Z}} \sum_{m \neq 0} \frac{1}{(mz+n)(mz+n+1)}.$$

Finally, consider the sum

$$\lim_{N \rightarrow \infty} \sum_{n=-N}^N \sum_{m \neq 0} \left(\frac{1}{mz+n} - \frac{1}{mz+n+1} \right).$$

For each fixed N we can reverse the sum and calculate, since the inner sum telescopes:

$$\begin{aligned} \sum_{m \neq 0} \sum_{n=-N}^N \left(\frac{1}{mz+n} - \frac{1}{mz+n+1} \right) &= \sum_{m \neq 0} \left(\frac{1}{mz-N} - \frac{1}{mz+N} \right) \\ &= \frac{1}{z} \sum_{m \neq 0} \left(\frac{1}{N/z+m} + \frac{1}{N/z-m} \right) \\ &= \frac{2\pi}{z} \cot(\pi N/z). \end{aligned}$$

Taking the sum as $N \rightarrow \infty$ and using that

$$\pi \cot(\pi N/z) = \pi i - 2\pi i \sum_{m=0}^{\infty} e^{2\pi i m N/z}$$

we get the formula:

$$\lim_{N \rightarrow \infty} \sum_{n=-N}^N \sum_{m \neq 0} \left(\frac{1}{mz+n} - \frac{1}{mz+n+1} \right) = \frac{2\pi i}{z}.$$

This finishes the proof.

1.7.1 The Dedekind-eta function

Define $\eta(z)$ as through the following product

$$\eta(z) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n).$$

The function $\eta(z)$ is holomorphic and non-vanishing on $\mathbb{H}$ (to check it, it is enough to check that $\sum q^n$ converges absolutely and uniformly on compact subsets of $\mathbb{H}$).

Theorem 1.7.2 The η function satisfies the following transformation property:

$$\eta(-1/z) = \sqrt{z/i} \eta(z).$$

Proof. Note that:

$$\begin{aligned} \frac{d}{dz} \log \eta(z) &= \frac{2\pi i}{24} + \sum_{n=1}^{\infty} \frac{2\pi i n q^n}{1 - q^n} = \frac{2\pi i}{24} \left(1 + 24 \sum_{n=1}^{\infty} \frac{n q^n}{1 - q^n} \right) \\ &= \frac{\pi i}{12} \left(1 + 24 \sum_{n=1}^{\infty} n \sum_{m=1}^{\infty} q^{nm} \right) = \frac{\pi i}{12} E_2(z). \end{aligned}$$

Using quasi-modularity of E_2 , we deduce:

$$d \log(\eta(-1/z)) = d \log(\sqrt{z/i} \eta(z)).$$

Therefore $\eta(-1/z) = C \sqrt{z/i} \eta(z)$ and setting $z = i$ one gets $C = 1$, as we wanted.

Theorem 1.7.3

$$\eta^{24}(z) = q \prod_{n=1}^{\infty} (1 - q^n)^{24}.$$

Proof. Note that:

$$\eta^{24}(z) = q \prod_{n=1}^{\infty} (1 - q^n)^{24},$$

so $\eta^{24}(z+1) = \eta^{24}(z)$. Moreover,

$$\eta^{24}(-1/z) = z^{12} \eta^{24}(z).$$

Since $\eta^{24}(z) = q + \dots$ and $\eta^{24}(z) \in S_{12} = \mathbb{C}$, we deduce that $\eta^{24} =$.

1.8 Growth of Fourier coefficients

In this final section we study the different behavior of the Fourier coefficients of a modular form f , depending on whether f is cuspidal or not. Intuitively, if f is cuspidal, the vanishing at infinity should force the coefficients to grow slower.

We start by studying Eisenstein series.

Proposition 1.8.1 The Fourier coefficients $a_n(E_k)$ of the Eisenstein series E_k grow as n^{k-1} . Precisely, there exist constant $A, B > 0$ such that

$$n^{k-1} \leq |a_n(E_k)| \leq B n^{k-1}.$$

Proof. We will use that the Fourier coefficients of E_k have a simple formula:

$$E_k = 1 + C_k \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n,$$

and hence $|a_n(E_k)| = |C_k| \sigma_{k-1}(n)$. Therefore

$$|a_n(E_k)| = |C_k| \sum_{d|n} d^{k-1} \geq |C_k| n^{k-1},$$

and we may take $B = |C_k|$. On the other hand,

$$\frac{|a_n(E_k)|}{n^{k-1}} = |C_k| \sum_{d|n} \left(\frac{d}{n}\right)^{k-1} = |C_k| \sum_{d|n} \frac{1}{d^{k-1}}.$$

This last term can be coarsely estimated:

$$|C_k| \sum_{d|n} \frac{1}{d^{k-1}} \geq |C_k| \sum_{d=1}^{\infty} \frac{1}{d^{k-1}} = |C_k| \zeta(k-1).$$

Hence we may take $B = |C_k| \zeta(k-1)$.

The following result shows that the coefficients of cusp forms grow much slower.

Theorem 1.8.2 — Hecke. If $\sum_{n=1}^{\infty} a_n q^n$ is the q -expansion of a cusp form of weight k , then $a_n = O(n^{k/2})$. Precisely, there is a constant $M > 0$ such that

$$|a_n| \leq M n^{k/2}.$$

Proof. Note that as $q \rightarrow \infty$ we have $|f(z)| = O(q) = O(e^{-2\pi i y})$, where $y = \Im(z)$. Consider the function

$$\phi(z) = |f(z)| y^{k/2}.$$

It is a continuous function $\mathbb{H} \rightarrow \mathbb{R}_{\geq 0}$, and it is invariant under $\mathrm{SL}_2(\mathbb{Z})$. Since ϕ approaches 0 as y approaches infinity, we deduce that ϕ is bounded: $|f(z)| \leq M' y^{-k/2}$ for all $z \in \mathbb{H}$. Since $f(z) q^{-n-1} = \dots + a_n q^{-1} + a_{n+1} + a_{n+2} q + \dots$ the residue theorem implies that

$$a_n = \frac{1}{2\pi i} \int_{C_y} f(z) q^{-n-1} dq,$$

where C_y is the counter-clockwise circle described by $e^{2\pi i(x+iy)}$ when y is fixed and x moves from 0 to 1. This gives:

$$a_n = \int_0^1 f(x+iy) q^{-n} dx,$$

Using the bound for $|f|$ we obtain:

$$|a_n| \leq \int_0^1 M' y^{-k/2} |e^{-2\pi i n(x+iy)}| dx = M' y^{-k/2} e^{2\pi n y}.$$

This expression is valid for all $y > 0$. In particular, for $y = 1/n$ we get $|a_n| \leq M' e^{2\pi} n^{k/2}$. Setting $M = M' e^{2\pi}$ finishes the proof.

Corollary 1.8.3 If f is not a cusp form, then the coefficients a_n grow as n^{k-1} .

Proof. If f is a modular form of weight k , write $f = \lambda E_k + h$, where h is a cusp form. The hypothesis of f not being a cusp form translates in $\lambda \neq 0$. Therefore

$$a_n(f) = \lambda a_n(E_k) + a_n(h).$$

Since n^{k-1} grows much faster than $n^{k/2}$ and $\lambda \neq 0$, we deduce that $a_n(f)$ grows as n^{k-1} .

2. Congruence Subgroups

2.1 Congruence subgroups

Let $N \geq 1$ be an integer. In this section we will consider subgroups of $\mathrm{SL}_2(\mathbb{Z})$ that are especially nice to work with. There are other subgroups that are interesting but beyond the scope of this course.

Definition 2.1.1 The *principal congruence subgroup* of level N is

$$\Gamma(N) = \{\gamma \in \mathrm{SL}_2(\mathbb{Z}) \mid \gamma \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N}\}.$$

Note that $\Gamma(1) = \mathrm{SL}_2(\mathbb{Z})$, so we are strictly generalizing Chapter 1. Note also that $\Gamma_0(N)$ can be defined as the kernel of the group homomorphism induced by the reduction map $\mathbb{Z} \rightarrow \mathbb{Z}/N\mathbb{Z}$:

$$\pi_N: \mathrm{SL}_2(\mathbb{Z}) \rightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}).$$

Therefore, $\Gamma(N)$ is a normal subgroup of $\mathrm{SL}_2(\mathbb{Z})$, of finite index.

Proposition 2.1.1 The map π_N is surjective.

Proof. Exercise.

There are too few principal congruence subgroups (only one for each $N \geq 1$), so it is desirable to consider more general subgroups.

Definition 2.1.2 A subgroup $\Gamma \subseteq \mathrm{SL}_2(\mathbb{Z})$ is a *congruence subgroup* if there is some $N \geq 1$ such that

$$\Gamma(N) \subseteq \Gamma \subseteq \mathrm{SL}_2(\mathbb{Z}).$$

The *level* of a congruence subgroup Γ is the minimum N such that $\Gamma(N) \subseteq \Gamma$.

One can think of many different ways to construct congruence subgroups. There are two families that arise so frequently that have special notation for them:

Example 2.1 For each $N \geq 1$, define

$$\Gamma_1(N) = \left\{ \begin{pmatrix} * & * \\ 0 & 1 \end{pmatrix} \pmod{N} \right\},$$

and also

$$\Gamma_0(N) = \left\{ \begin{pmatrix} * & * \\ 0 & * \end{pmatrix} \pmod{N} \right\}.$$